

ECE 601 – Machine Learning for Engineers

Spring 2026 Course Project – Phase 1: Proposal

University of Massachusetts Amherst

April 2, 2026

(a) Project Description

Problem Overview

The goal of this project is to build a multi-class image classifier that can automatically identify the type of skin lesion present in a dermatoscopic photograph. Accurate and automated skin lesion classification is a clinically significant problem: melanoma, the deadliest form of skin cancer, is highly curable when caught early, yet visual diagnosis is error-prone even among trained dermatologists. A reliable machine learning classifier trained on standardized dermoscopy images could serve as a decision-support tool in clinical settings.

(i) Input Data

The dataset used for this project is the **HAM10000** (*Human Against Machine with 10,000 training images*) dataset, a widely used benchmark in medical image analysis. It is publicly available on Kaggle at:

<https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

The dataset contains **10,015 dermatoscopic images** in JPEG format, each depicting a pigmented skin lesion on a patient's skin surface. Images were collected from multiple institutions across Austria and Australia and span a diverse range of patient ages, skin types, and body locations. Accompanying metadata is provided in a CSV file and includes the following fields:

- `lesion_id`: unique identifier for each lesion
- `image_id`: unique identifier for each image
- `dx`: ground-truth diagnosis label (one of 7 classes, described below)
- `dx_type`: method of diagnosis confirmation (histopathology, follow-up, expert consensus, or confocal microscopy)
- `age, sex, localization`: patient demographic and lesion location metadata

Images can be downloaded directly from Kaggle using the Kaggle API (`kaggle datasets download`) or manually via the Kaggle web interface. No additional data collection is required.

(ii) Learning Target

The target variable is the ground-truth diagnostic class `dx`, which takes one of the following **7 values**:

- `mel` – Melanoma
- `nv` – Melanocytic Nevi (benign moles)
- `bcc` – Basal Cell Carcinoma
- `akiec` – Actinic Keratoses and Intraepithelial Carcinoma
- `bkl` – Benign Keratosis-like Lesions
- `df` – Dermatofibroma
- `vasc` – Vascular Lesions

This is a **supervised multi-class classification** problem. The input to the model is a raw RGB image (resized to a fixed resolution, e.g., 224×224 pixels), and the output is a probability distribution over the 7 diagnostic classes, from which the predicted class is taken as the $\arg \max$. The dataset exhibits significant class imbalance (approximately 67% of images belong to the `nv` class), which will be an important consideration in both training and evaluation. Performance will be assessed using overall accuracy as well as balanced/macro-averaged metrics (macro F1-score, per-class recall) to account for the class imbalance.

(b) AI Coding Assistant Prompt

The following prompt will be submitted verbatim to an AI coding assistant (e.g., Claude, GitHub Copilot, or OpenAI Codex) to obtain the initial implementation:

Please write a complete Python script that:

1. Loads and preprocesses the images (resize to 224×224 , normalize pixel values)
2. Splits the data into training, validation, and test sets
3. Trains a convolutional neural network to classify each image into one of the 7 classes
4. Evaluates performance on the test set using accuracy and macro-averaged F1-score
5. Plots the training and validation loss and accuracy curves

Use PyTorch for the implementation and target a standard GPU (NVIDIA RTX 3060). The script should be runnable end-to-end without requiring any changes. *I have a medical image classification problem. I have a dataset of approximately 10,000 dermoscopic skin lesion images, each belonging to one of 7 diagnostic categories: melanoma, melanocytic nevi, basal cell carcinoma, actinic keratoses, benign keratosis-like lesions, dermatofibroma, and vascular lesions. The images are in JPEG format and labels are provided in a CSV file with a column called dx. The dataset has significant class imbalance, with the majority of images being melanocytic nevi.*

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